



The 4th International Conference of Tropical Plants (ICTP2026)

22 – 25 March 2026

The Zign hotel· Pattaya Chonburi · Thailand

<https://bioinformatics.hainanu.edu.cn/ictp4/>

Artificial Intelligence and Digital Technologies for Advancing Crop Research and Breeding: A Review

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CIRAD, UMR AGAP Institut

4th International Conference of Tropical Plants (ICTP2026)

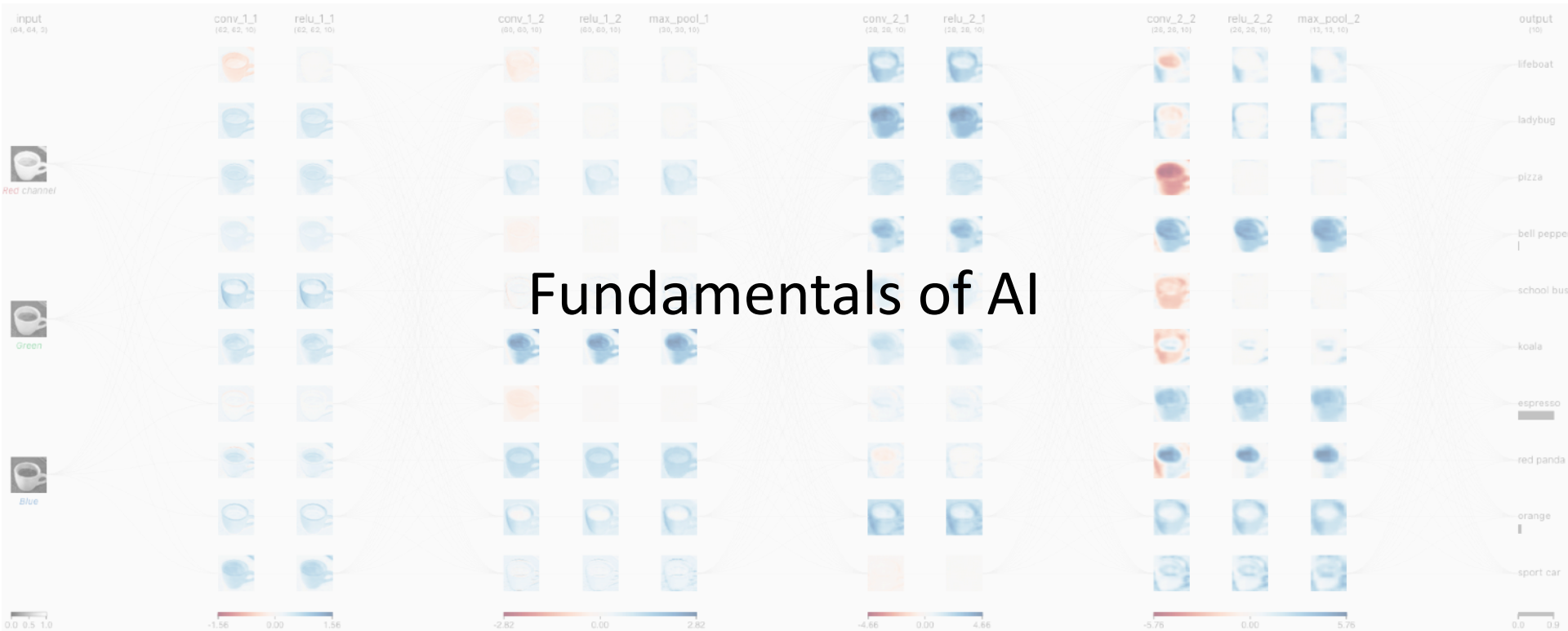
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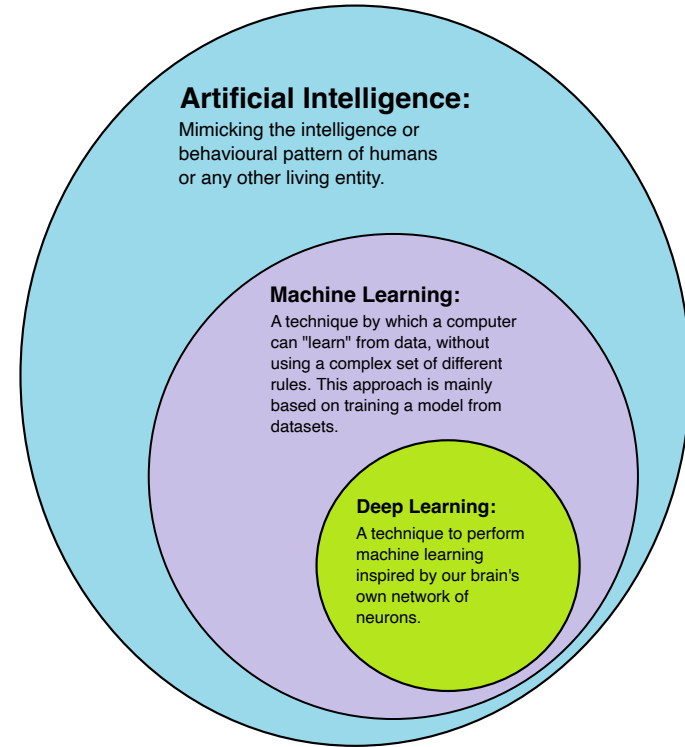
Outline

- Fundamentals of AI
- Examples of AI applications in agriculture, characterization and selection of materials
- Perspectives on utilizing AI to accelerate coconut research



Defining AI, ML and DL

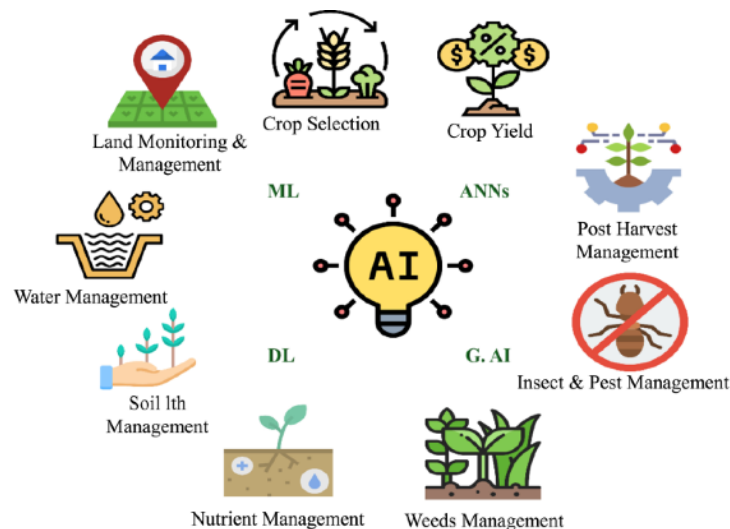
- **Artificial Intelligence (AI):** Computational systems capable of performing tasks that typically require human intelligence:
 - pattern recognition
 - decision-making
 - learning
- **Machine Learning (ML):** Algorithms that learn from data to make **predictions** or **decisions** without being explicitly programmed.
- **Deep Learning (DL):** Uses artificial neural networks to automatically extract complex features from large datasets.



<https://commons.wikimedia.org>

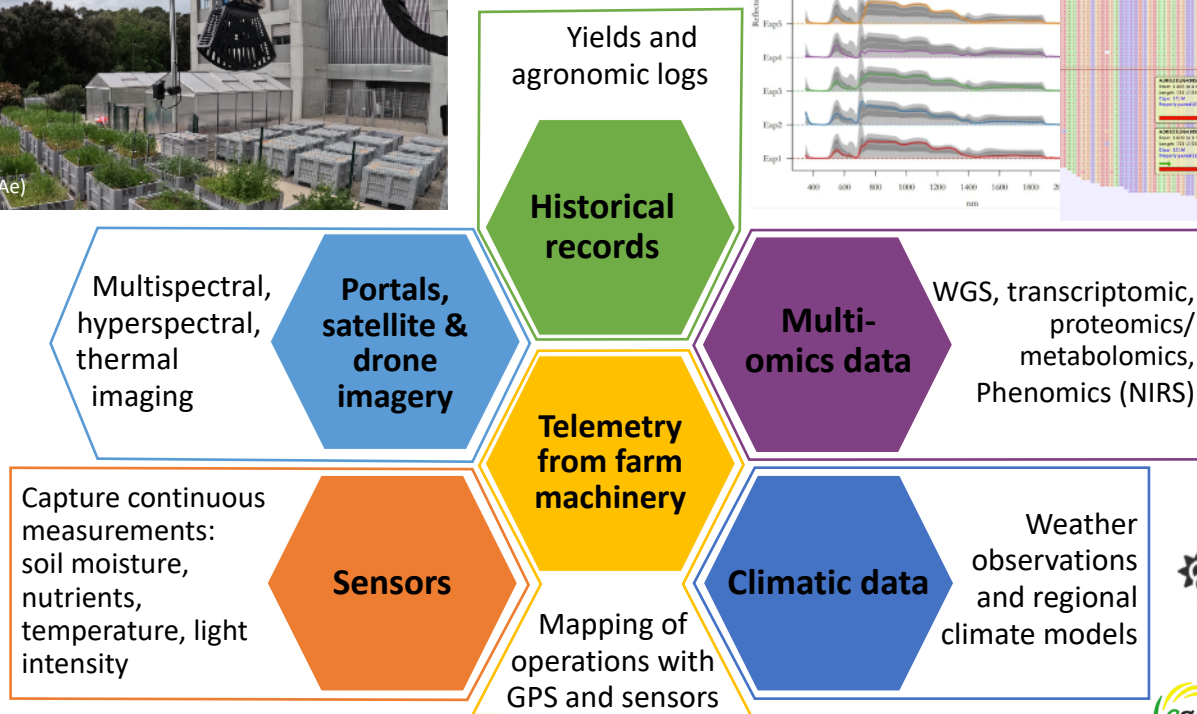
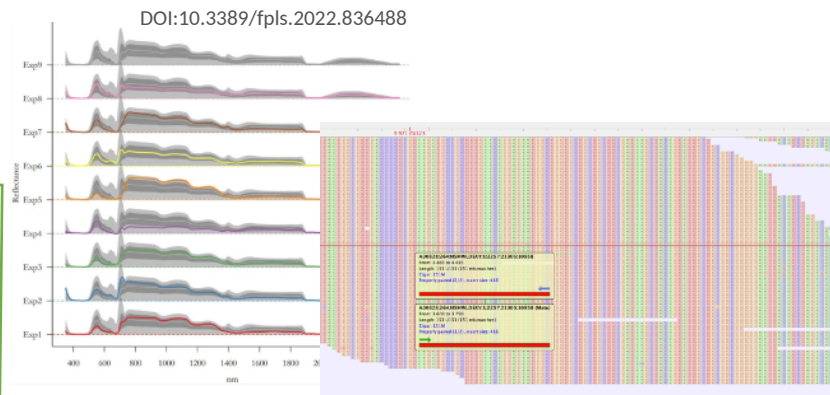
Why AI, ML and DL in agriculture?

- Modern agriculture faces major challenges (climate change, growing food demand, and resource constraints) that require data-driven and scalable solutions.
- **Precision agriculture**, or digital agriculture, depends on **big data**, **sensors**, and **high-performance computing** to improve decision-making and sustainability.
- Models can analyze diverse data sources to:
 - **Predict** yields and optimize inputs
 - **Detect** pests and diseases early through image-based diagnostics
 - **Monitor in real time** soil health and water usage



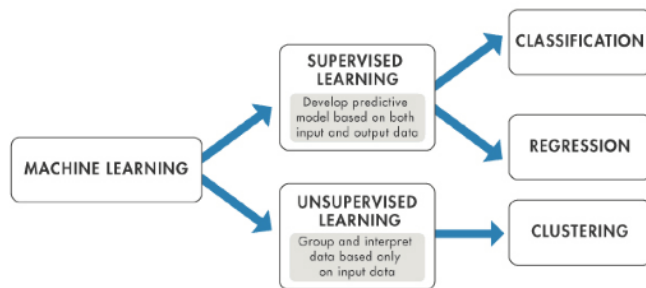
DOI:10.3389/fpls.2022.836488

Fundamentals: Data sources in agriculture



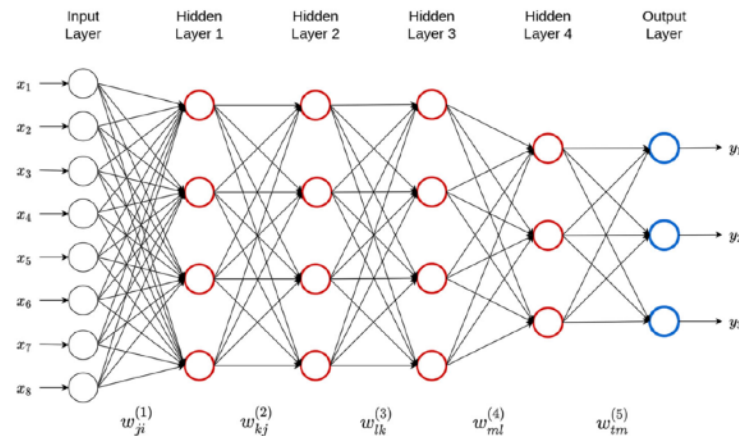
Fundamentals: Supervised / Unsupervised learning

- Learning approaches:
 - Supervised:** Models are trained using labelled examples (categorical, real value)
 - Unsupervised:** Models discover patterns without labels (clustering)
- Algorithms and architectures



<https://www.datacamp.com/blog/what-is-machine-learning>

Term	Description
Traditional ML	Algorithms like Random Forests, SVM, k-NN or G-BLUP learn from structured data to infer traits or classify conditions. Models are relatively transparent and require feature extraction.
Deep Learning	Neural network–based architectures (CNN, RNN, Transformers) that automatically learn hierarchical features directly from raw images or data.

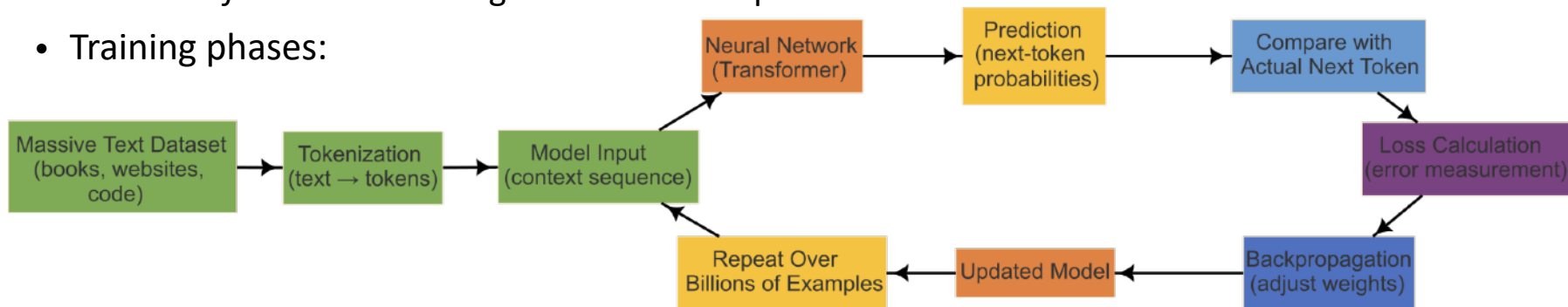


DOI:10.1186/s12864-020-07319-x

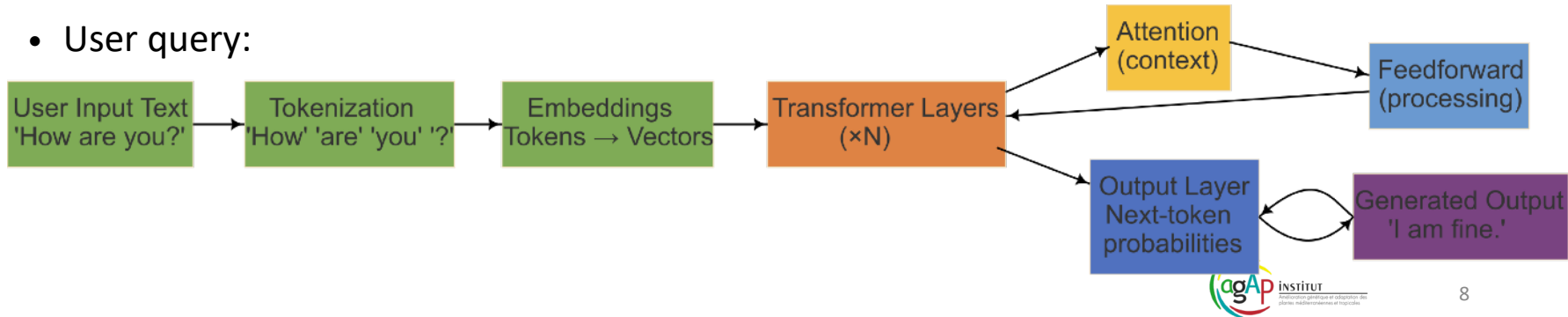
Fundamentals: Large Language Models (LLMs)

LLMs are AI systems trained on massive text datasets to understand and generate human-like language.

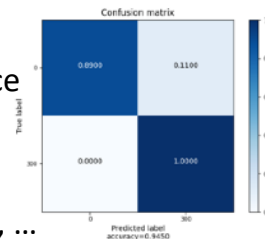
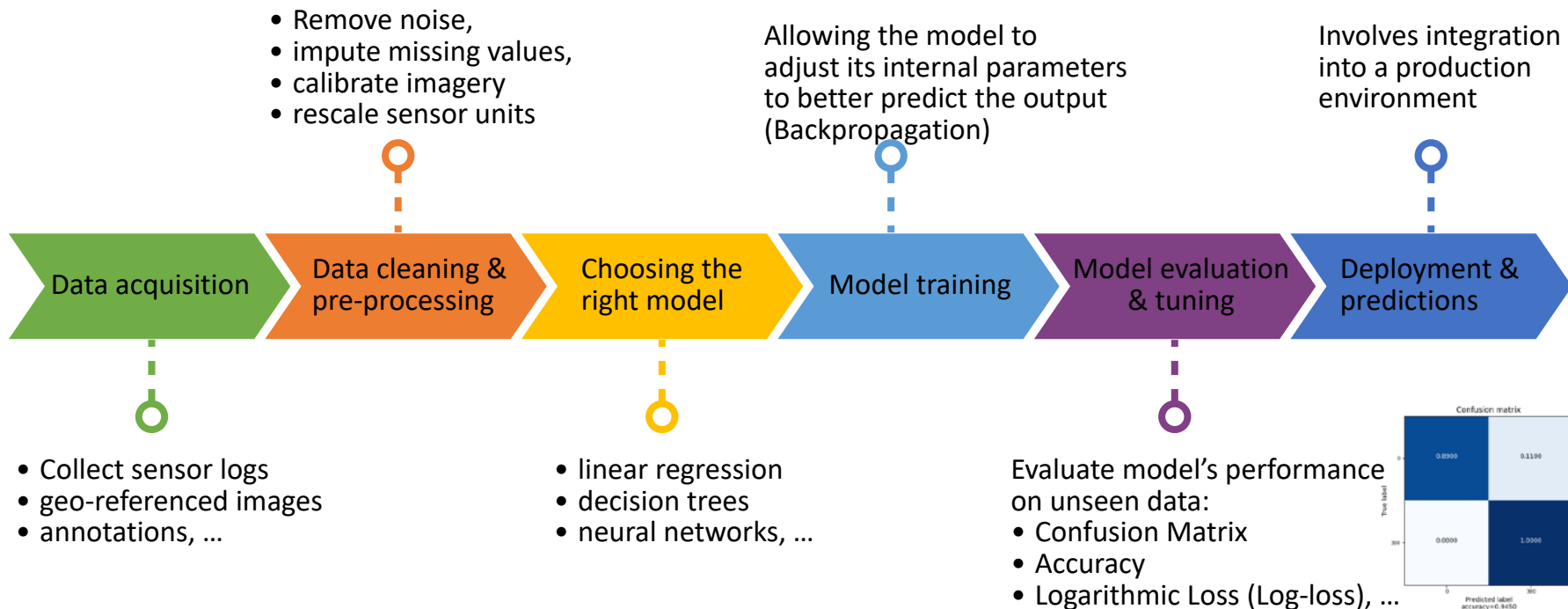
- Predict the next token (word/subword) based on context
- Uses *self-attention* to weigh the relationships between words
- Training phases:



- User query:

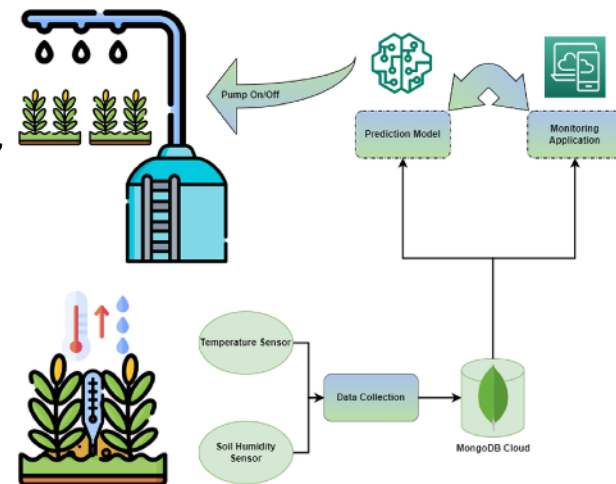


Fundamentals: Typical Workflow for AI/ML Pipelines

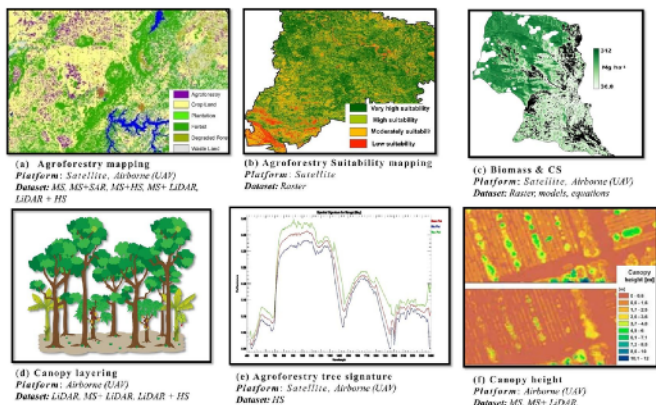


AI applications in agriculture

- Water management:
 - **AI-based smart irrigation:** monitoring the status of soil water, crop growth conditions, and temporal and spatial patterns in combination with weather conditions monitoring and forecasting to efficiently manage water.
- Soil management:
 - Remote sensing and soil mapping sensors.



DOI:[10.1016/j.egy.2022.07.088](https://doi.org/10.1016/j.egy.2022.07.088)



DOI:[10.1007/s11356-022-20305-y](https://doi.org/10.1007/s11356-022-20305-y)

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AI applications in agriculture: Crop management

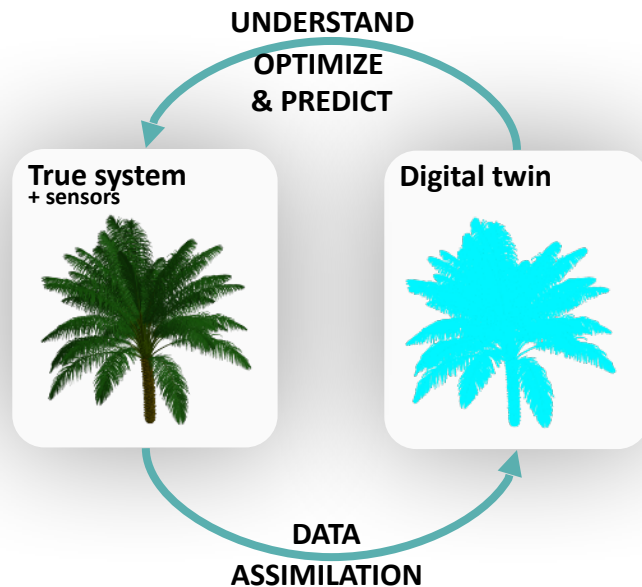
- **Crop quality:** Developing decision support systems can aid farmers in taking appropriate management decisions for increased quality of production.
- **Yield prediction:** Models that integrate soil, weather, crop biomass, phenology, water and nutrient uptake, and historical yield data to forecast production.

Digital twin: Virtual model to accurately reflect a physical object or system and its dynamic.

- **Challenge:** Complex Systems

- Understand:
 - Isolate, test and validate hypotheses for each process
 - Identify the most suitable biological or physical models.
- Predict:
 - Tasks, **yield**, and **risks** under different management scenarios
- Optimize:
 - Find the optimal plant traits and management

→ Tailored to each plant and local conditions

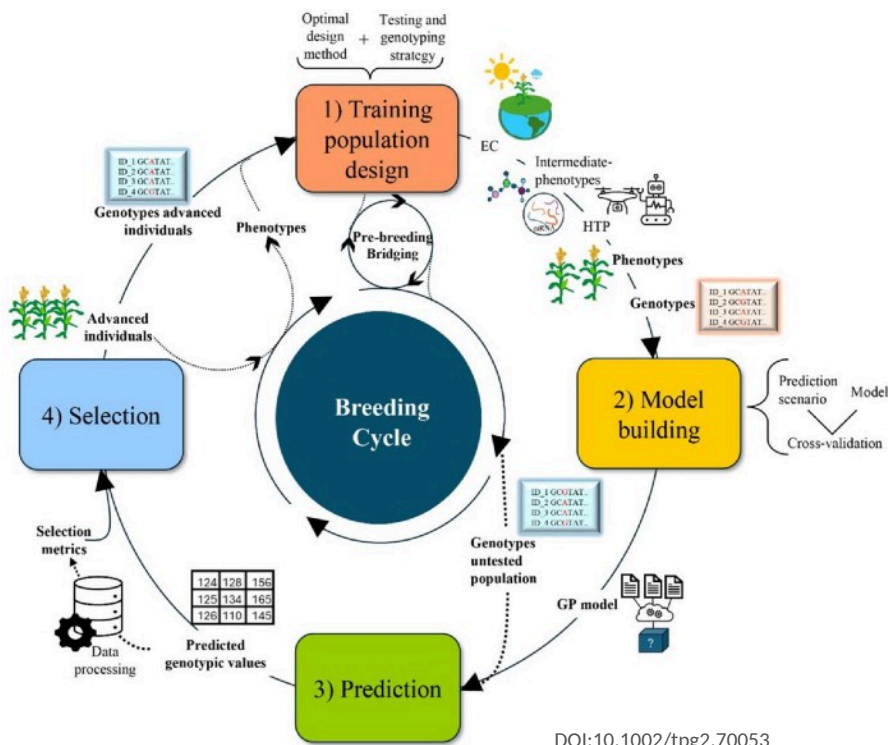


From a presentation by R. Vezy

Machine learning in plant breeding: Genomic Selection (GS)

- **GS** (or Genomic Prediction) uses genome-wide genetic markers to estimate **Genomic Estimated Breeding Values (GEBVs)** of individuals, enabling early selection without phenotype measurement.
- GS estimates effects across all markers, in contrast with marker-assisted selection (MAS), which focuses on a few significant markers
- Aims to accelerate breeding by enabling earlier selection and shorter cycles.

Typical workflow:



DOI:10.1002/tpg2.70053



Methods used in GS

- Best Linear Unbiased Prediction (GBLUP): Models breeding values with mixed models.
- Bayesian approaches: Bayesian Ridge Regression, Bayes A, Bayes B, etc.
- ML models: Support Vector Regression (SVR), Random Forests (RF).
- DL models: multilayer artificial neural networks (DNNs, CNNs, MLPs).

GBLUP	DL
Competitive for traits with simpler structures dominated by additive genetics effects	Excels in traits involving complex relationships, by capturing non-linear patterns

* Based on a comparison across 14 datasets analyzed

DOI:10.3389/fgene.2025.1568705

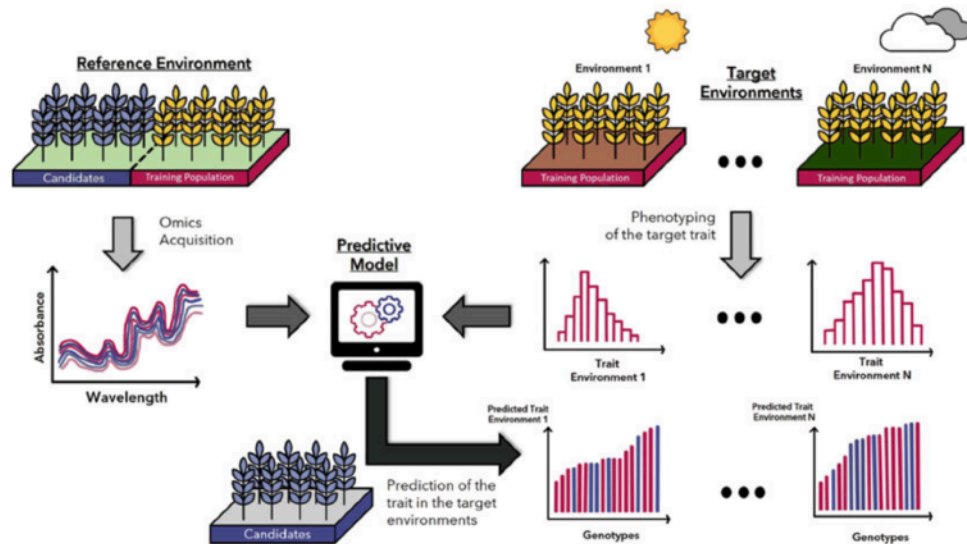
Model choice should depend on trait characteristics and the breeding program's evaluation priorities.

Success of DL models significantly depended on careful parameter optimization

Machine learning in plant breeding: Phenomic Selection (PS)

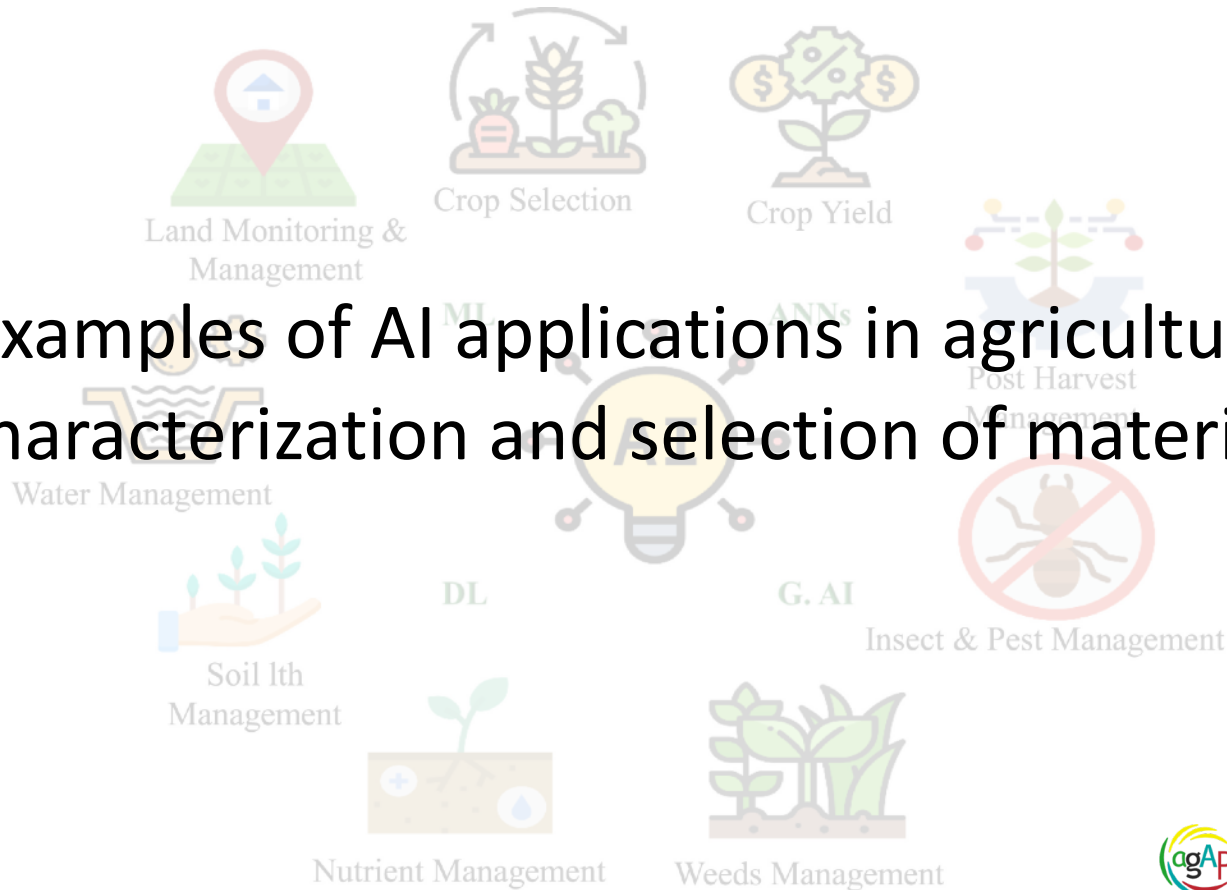
- **PS** is a high-throughput, low-cost method that replaces genotyping with **phenomic data** (like NIRS), to construct relationship matrices for **predicting** complex traits.
- PS uses NIRS as genetic **proxies**, capturing broader genetic relationships rather than immediate trait values.
- Offers a low-cost, high-throughput, and non-destructive alternative to traditional genotyping.
- PS on parental NIR spectra can accurately predict hybrid performance
- Comparable or even superior predictive performance, and valuable flexibility across tissues and environments

Typical workflow:





Examples of AI applications in agriculture, characterization and selection of materials



DOI:10.3389/fpls.2022.836488

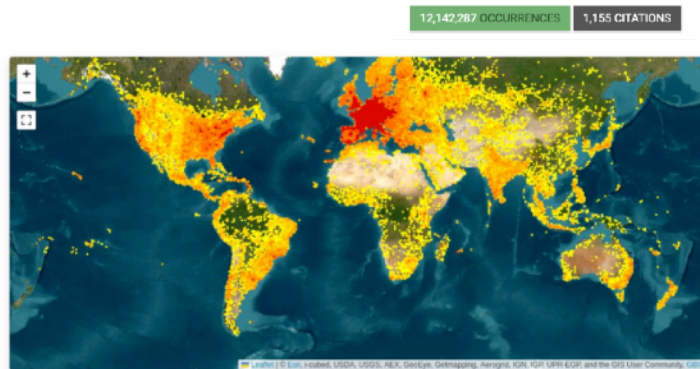
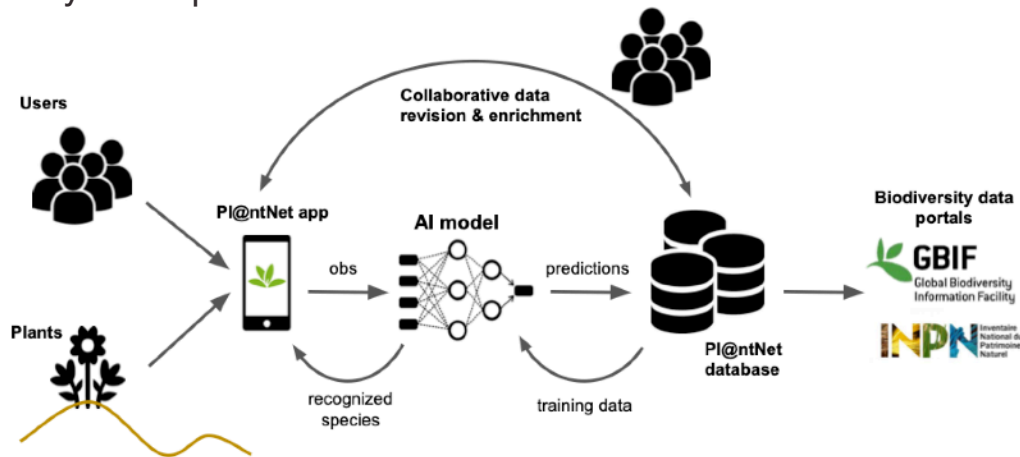
Plant Recognition:



Based on a presentation by the PI@ntNet team

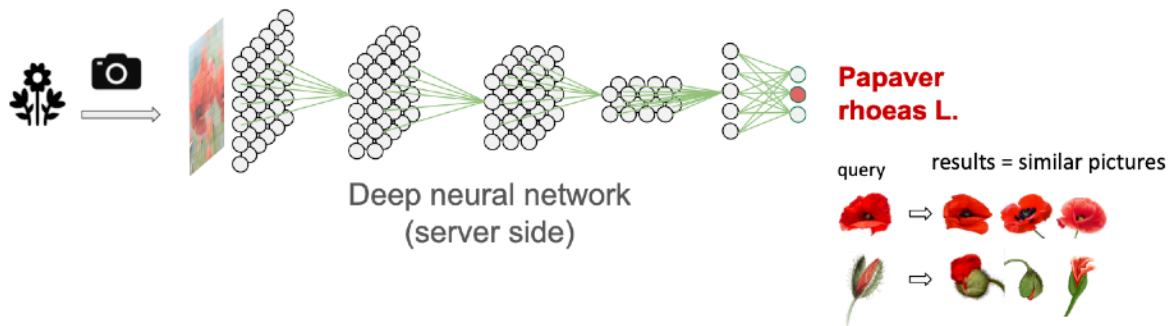
Key concept of PI@ntNet: Collaborative AI

Among the top plant data provider to GBIF



<https://doi.org/10.15468/mma2ec>

Classification (78K species) + Similarity search (16M images)



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PI@ntAgroEco



GENCI

UNIVERSITÉ DE MONTPELLIER

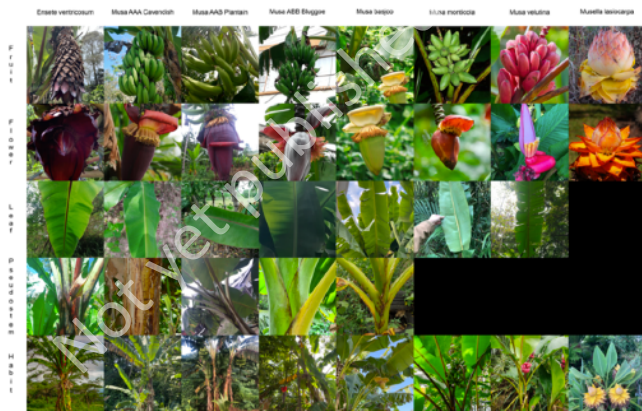
From image-based plant species recognition to varietal recognition

Varietal identification supports the selection and cultivation of crops best suited to local environmental contexts.



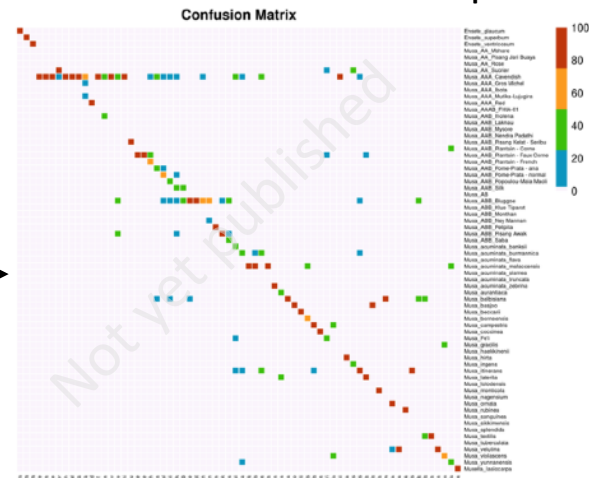
- Automating this task requires robust datasets.
- From an image analysis perspective, varietal recognition is an especially difficult task, involving very fine-grained classification that challenges the current limits of deep learning approaches.

The banana example:



4433 observations, 7150 images
in situ observations (citizen science) + **expert curation**

Trained
Model



→ Organ-level analysis revealed that specific plant features (fruits) are critical for effective classification.

Digital Twins: The PalmStudio project

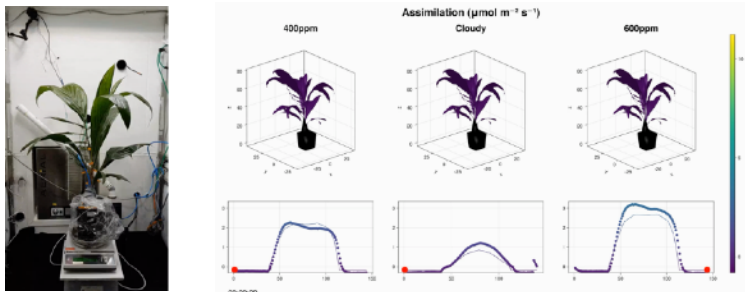
Adapted from a presentation by R. Vezy



Digital Twin 1: Biophysics

Objective: Validate and compare hypotheses for biophysical processes

→ Simulation of plant response to climate.

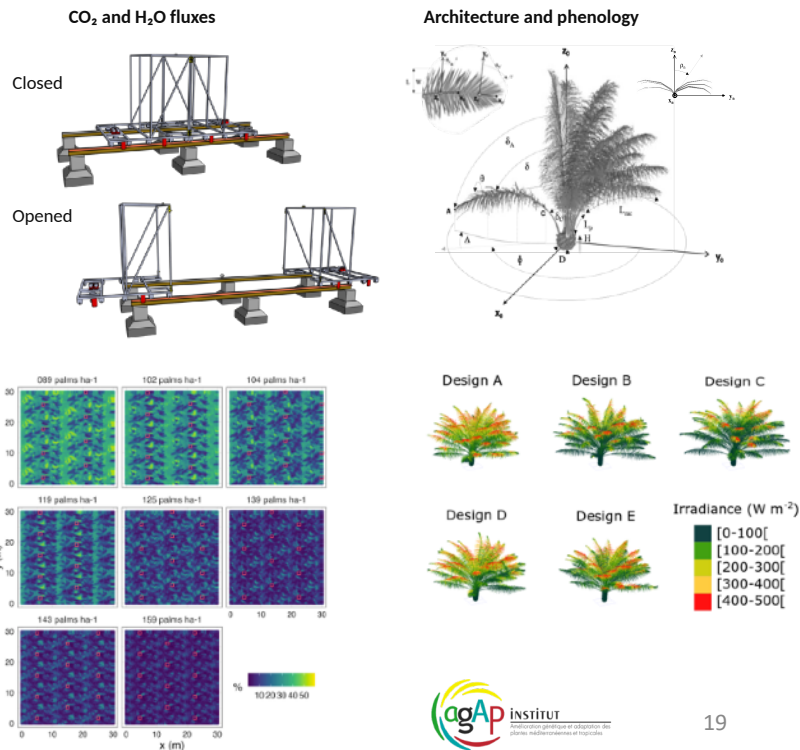


Broad Applications:

- Guide management:
 - Innovative designs: density effect, agroforestry (Perez et al. 2022, 2024)
 - Optimized inputs (fertilizers, irrigation...)
- Assist genetic selection (ideotyping) (Perez et al. 2018)
- Predict impact of El Niño, haze, climate changes...
- Carbon footprint evaluation, data for certification standards
- Reduce manpower (e.g. robotics)?

Digital Twin 2: Growth

Objective: Evaluate models related to growth under real field conditions

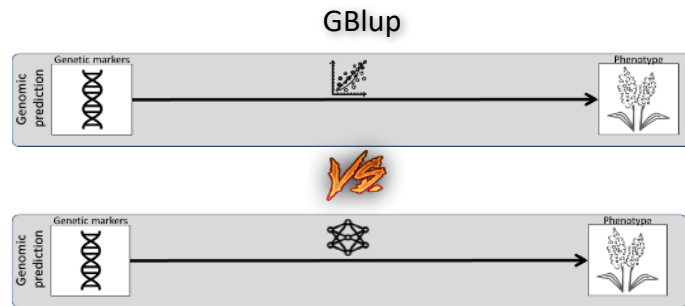




Using ML & DL for Genomic selection (in progress)

Oil Palm tree genomic selection

(SelGen_Palm)



Non linear domination

Machine Learning methods (fine-tuned)

RF	CatBoost	SVR	ElasticNet	XGBoost	Lasso	AdaBoost	KNN	GBlup	PLSRegression	Ridge
0.582	0.542	0.512	0.501	0.478	0.468	0.452	0.425	0.41	0.408	0.391

Deep Learning architectures

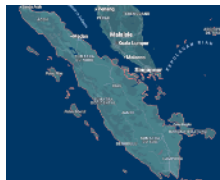
MLP	CNN (± Depthwise)	GRU	Transformer (encoder)	Transformer (encoder + CNN)	Transformer (complete)
0.561	0.545	0.554	0.470	0.536	0.553

85% > GBlup
 65% > 0.5

From a presentation by G. Beurier

Using ML & DL for Genomic selection (in progress)

Context:



Calibration

Site 1:
25520 palms/ 382 crosses



Evaluation

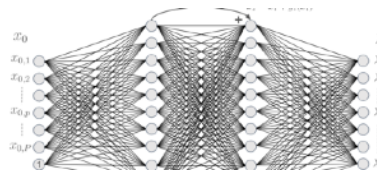
Site 2 (test set) :
536 crosses

Methods

X



SNPs: (~23000)

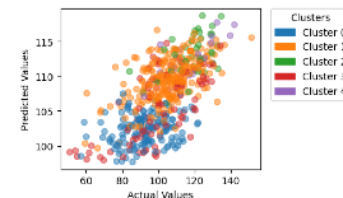


MLP/CNN/RNN/Transformer

Y

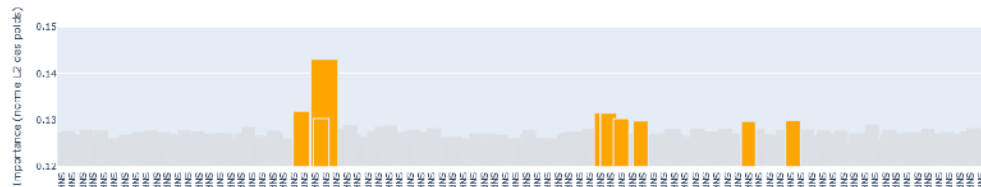
Prédiction :

- *FFB (Fresh Fruit Bunch)*
- *BN (Bunch Number)*
- *BW (Bunch Weight)*
- *HI (Height Increment)*



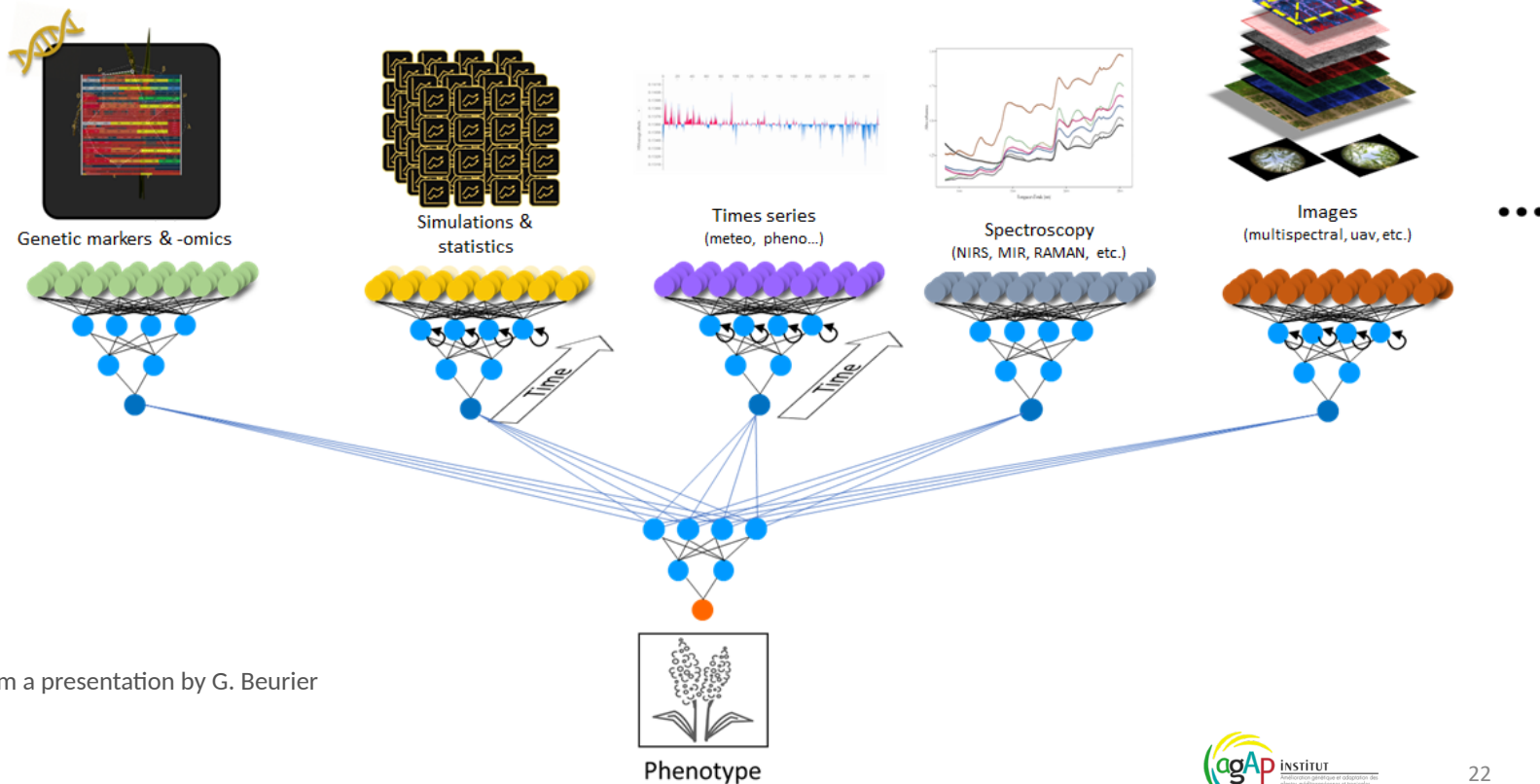
Explainability

Identify SNPs linked to traits of interest



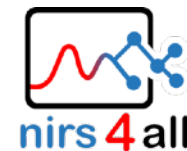
Multi / Omni-modal architectures (in progress)

(NIRS + timeseries, -omics + timeseries, NIRS vs -omics, etc.)

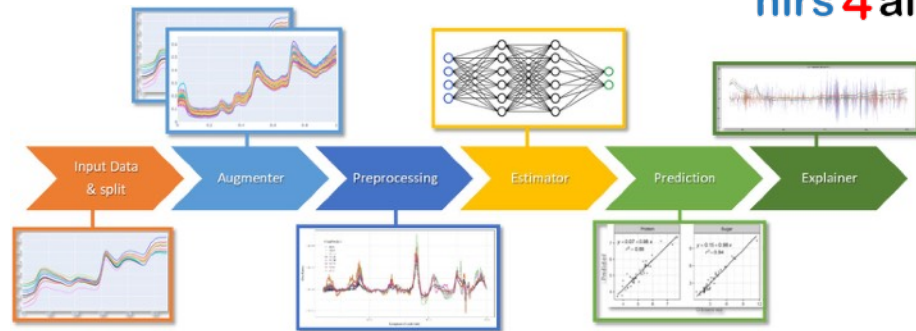


From a presentation by G. Beurier

Phenomics: NIRS4All is all you NIRS



- A near infrared spectroscopy ultra-complete pipeline
- numerous "shortcuts" allowing use without prior knowledge of NIRS or python
- a large number of unique, optimized algorithms for near infrared spectroscopy (in addition to classic methods)
- an open-source and free library



<https://pypi.org/project/nirs4all/>

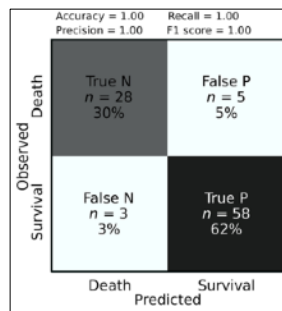
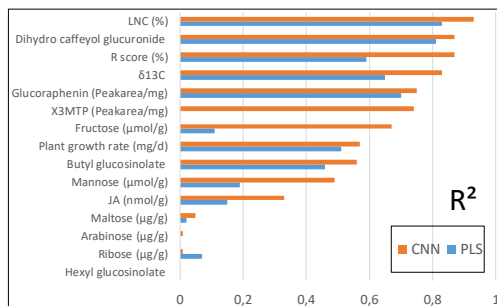


<https://github.com/GBeurier/nirs4all>

Regression

AND/OR

classification



- No more need to: find outliers, manage train/test splits, choose amongst preprocessing, select or tune a model
- AI non-linearity allows finding relationships that PLS cannot
- Explainability remains possible
- Switching from regression to classification
- Already validated and used by:

Vasseur et al. 2022 – Frontiers in Plant Science. DOI:10.3389/fpls.2022.836488

From a presentation by D. Cornet

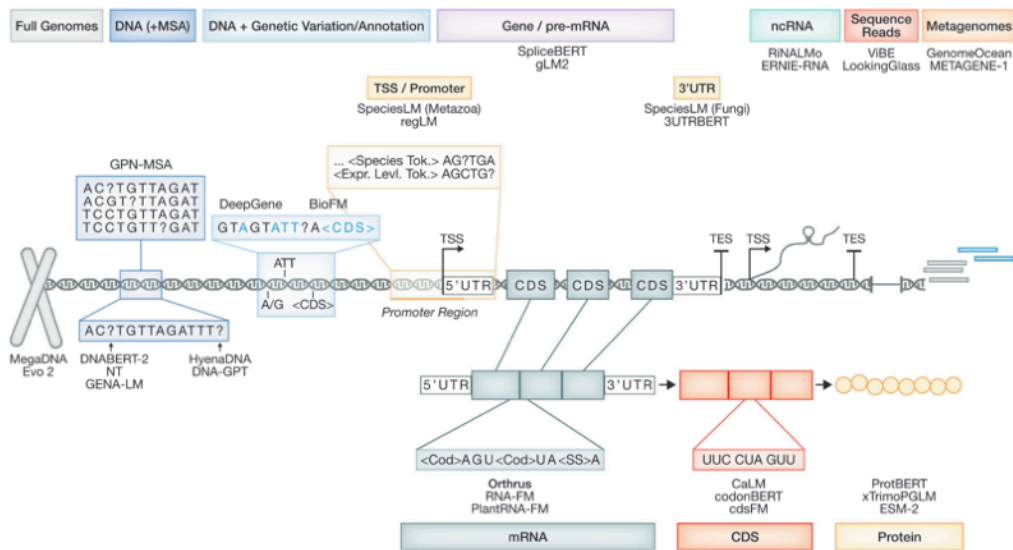
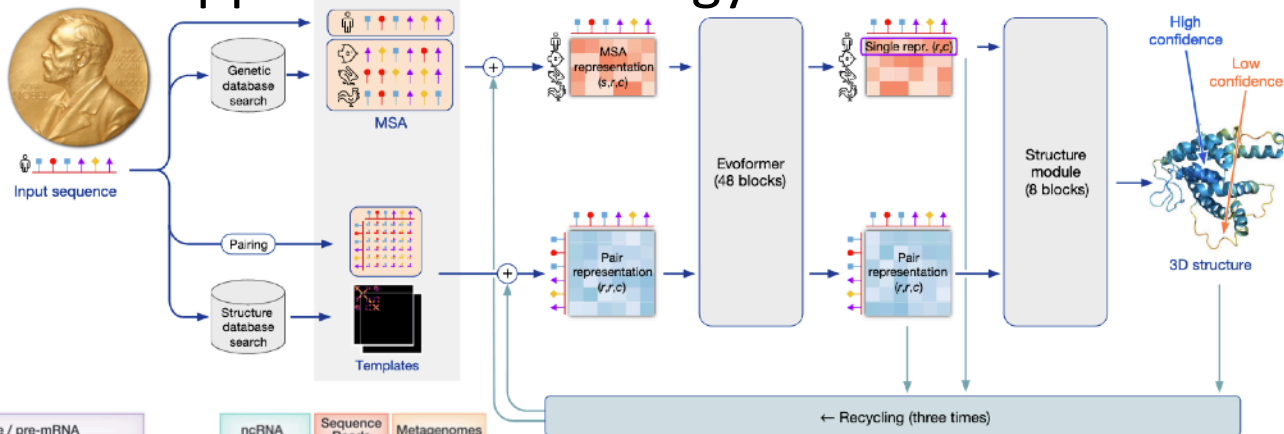


Advances that expand AI applications to biology

DOI:10.1038/s41586-021-03819-2

Protein models:

AlphaFold predicts the 3-D structure of proteins from their amino-acid sequence.



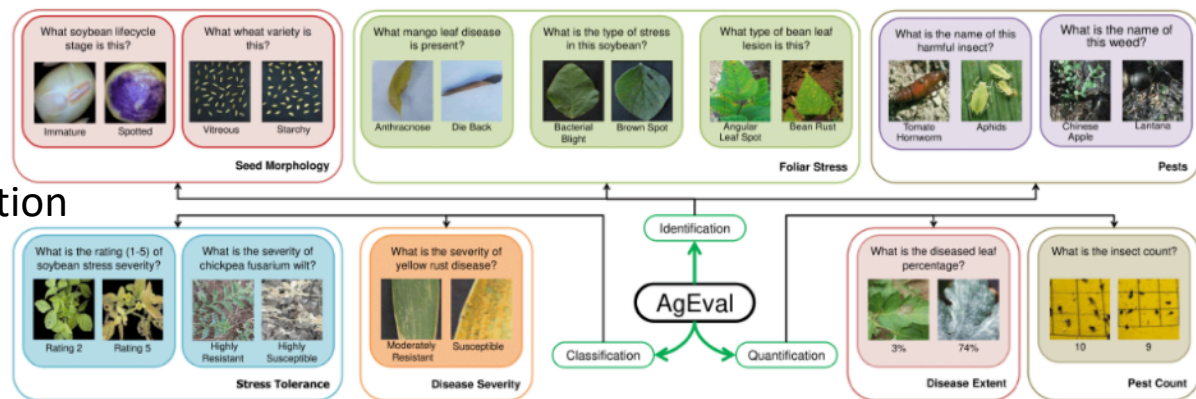
Pretrained Genomic language models:
Trained on different data

DOI:10.1038/s44320-025-00184-4

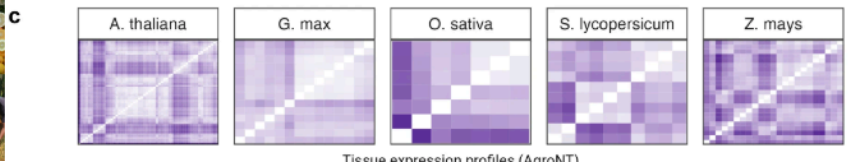
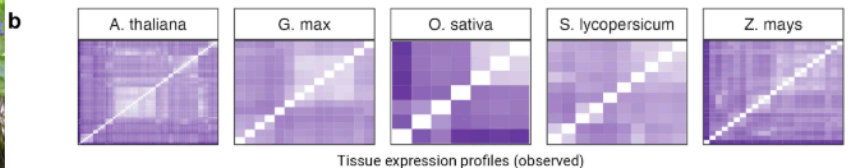
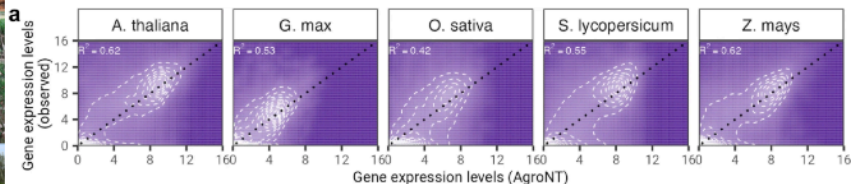
Advances in LLMs expand AI applications to plant breeding

• Plant phenotyping:

AgEval provides standardized tasks to evaluate multimodal AI models in plant stress detection



<https://github.com/arbab-ml/AgEval>



Modeling plant traits: AgroNT: trained to predict regulatory annotations

DOI:10.1038/s42003-024-06465-2



Perspectives on utilizing AI to accelerate coconut research

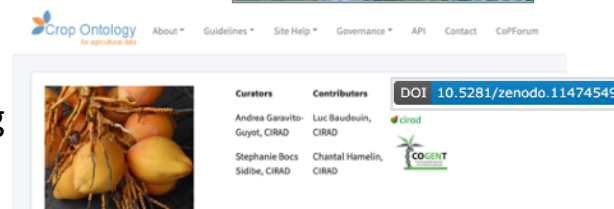




Upcoming coconut datasets to support AI-based approaches

Project **COCODIV** (Coconut diversity and Innovation Networking in the Pacific and South East Asia), financed by the Australian Centre for International Agricultural Research (**ACIAR**) will generate data from:

- Accessions present in ICGs and NCGs.
- Individuals from farmers' fields.
- Uncommon individuals identified during prospections in the coastal territories, public gardens and lands.
- Characterization data that will be **collected** and **curated** during COCODIV will include:
 - Passport data (**new** and **historical** records).
 - Local and vernacular nomenclatures, and socio-cultural value.
 - Genotyping (Whole genome low-coverage sequencing).
 - Morphological data (Traditional and new descriptors), following the Coconut Ontology.
 - Simple biochemical characterizations and NIRS measures from leaves, kernel and water.
- Climatic records from the original sampling sites can be consulted.



https://cropontology.org/term/CO_326:ROOT



Potential insights from applying AI to coconut data

Genetic diversity & Genomics:

- Is it feasible to detect fine-scale or hidden phenotypic and genetic structures across samples and regions
- Can NIRS profiles be used to reveal coconut's genetic structure?
- Can morphotypes and varieties be identified automatically?

Trait Prediction:

- Can NIRS be used to predict biochemical or morphological traits?
- Can historical yield records combined with genotypic data be used to predict yield or trait performance, through genomic selection?
- Can resequencing data, morphological descriptors, and NIRS features be integrated to develop multi-modal prediction models?

Adaptation:

- Can climatic data from origin sites, combined with genotypic and morphological information, reveal genomic patterns of local adaptation and help identify potentially well-adapted materials?
- Can accessions with potential adaptation to particular climate or soil environments be identified?



Takeaway message

- **AI** can unlock the **value of existing collections** by analyzing genomic, morphological, and NIRS data to identify elite traits and guide more **informed breeding decisions**.
- Integrating diverse datasets with machine learning enables faster, more **cost-effective selection pipelines**, especially for long-cycle crops like coconut.
- **Digital tools** support conservation by helping to cluster, compare, and prioritize accessions for *in situ* and *ex situ* strategies based on **genetic and adaptive diversity**.

Note:* Investing in **capacity building and **data infrastructure** is essential to ensure that coconut-producing countries can fully benefit from AI-powered technologies in breeding and conservation.

Acknowledgements

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Théo Simoes

Jean-Marc Sadaillan

Christophe Botella

Charlotte Tibi

Yves Caraglio

Jean-Christophe Lombardo

Joseph Salmon

* I'm just the messenger!





Dziękuję घन्ययद Merci

Gracias ありがとうございます

Danke Obrigado Grazie

Terima kasih **Thank You**

நன்றி Teşekkürler

Bahot shukriyaa 감사합니다